17	investigate and calculate power from the rate at which work is done or energy transferred	Estimate power output of electric motor (see also outcome 53)

1.4 Materials

This topic covers flow of liquids, viscosity, Stokes' law, properties of materials, Hooke's law, Young's modulus and elastic strain energy.

This topic may be taught using, for example, a case study of the production of sweets and biscuits. It could also be taught using the physics associated with spare part surgery for joint replacements and lens implants.

Learning outcomes 18–27 should be studied using variety of applications, for example, making and testing food, engineering materials, spare part surgery. This unit includes many opportunities to develop experimental skills and techniques.

Students will be assessed on their ability to:	Suggested experiments
18 understand and use the terms *density*, *laminar flow*, *streamline flow*, *terminal velocity*, *turbulent flow*, *upthrust* and *viscous drag*, for example, in transport design or in manufacturing	
19 recall, and use primary or secondary data to show that the rate of flow of a fluid is related to its viscosity	
20 recognise and use the expression for Stokes's Law, $F = 6\pi\eta rv$ and upthrust = weight of fluid displaced	
21 investigate, using primary or secondary data, and recall that the viscosities of most fluids change with temperature. Explain the importance of this for industrial applications	
22 obtain and draw force–extension, force–compression, and tensile/compressive stress-strain graphs. Identify the *limit of proportionality*, *elastic limit* and *yield point*	Obtain graphs for, example, copper wire, nylon and rubber

Students will be assessed on their ability to:	Suggested experiments
23 investigate and use Hooke's law, $F = k\Delta x$, and know that it applies only to some materials	
24 explain the meaning and use of, and calculate *tensile/compressive stress*, *tensile/compressive strain*, *strength*, *breaking stress*, *stiffness* and *Young Modulus*. Obtain the Young modulus for a material	Investigations could include, for example, copper and rubber
25 investigate elastic and plastic deformation of a material and distinguish between them	
26 explore and explain what is meant by the terms *brittle*, *ductile*, *hard*, *malleable*, *stiff* and *tough*. Use these terms, give examples of materials exhibiting such properties and explain how these properties are used in a variety of applications, for example, safety clothing, foodstuffs	
27 calculate the elastic strain energy E_{el} in a deformed material sample, using the expression $E_{el} = ½ Fx$, and from the area under its force/extension graph	